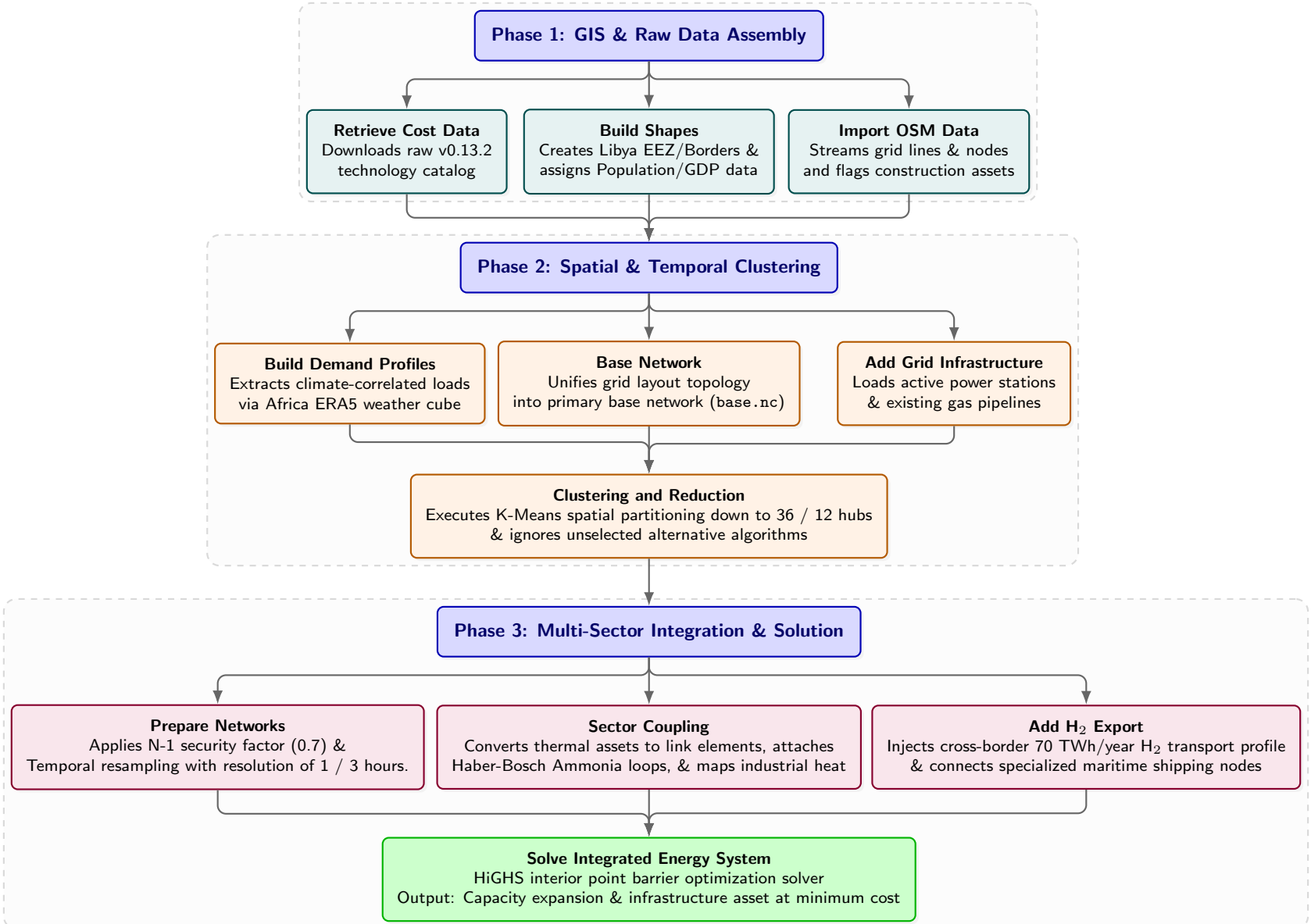
Phase 1: GIS & Raw Data Assembly
Retrieve Cost Data
Downloads raw v0.13.2 technology catalog
Build Shapes
Creates Libya EEZ/Borders & assigns Population/GDP data
Import OSM Data
Streams grid lines & nodes and flags construction assets
Phase 2: Spatial & Temporal Clustering
Build Demand Profiles
Extracts climate-correlated loads via Africa ERA5 weather cube
Base Network
Unifies grid layout topology into primary base network (base.nc)
Add Grid Infrastructure
Loads active power stations & existing gas pipelines
Clustering and Reduction
Executes K-Means spatial partitioning down to 36 / 12 hubs & ignores unselected alternative algorithms
Phase 3: Multi-Sector Integration & Solution
Prepare Networks
Applies N-1 security factor (0.7) & Temporal resampling with resolution of 1 / 3 hours.
Sector Coupling
Converts thermal assets to link elements, attaches Haber-Bosch Ammonia loops, & maps industrial heat
Add H_2 Export
Injects cross-border 70 TWh/year H_2 transport profile & connects specialized maritime shipping nodes
Solve Integrated Energy System
HiGHS interior point barrier optimization solver
Output: Capacity expansion & infrastructure asset at minimum cost